UNIT - 1 (One Variable Calculus)

- **Limits** - The limit of a function describes the value that the function approaches as the independent variable approaches a particular value.

$\lim_{x \to a} f(x) = L$ } means x gets closer & close to a, $f(x)$ gets closer & closer to L.

Imp point ⇒ $f(x) = \frac{x^2-4}{x-2}$ Here at $x = 2$, $f(x)$ is undefined.

But $= \frac{x^2-(2)^2}{x-2} \Rightarrow \frac{\cancel{(x-2)}(x+2)}{\cancel{(x-2)}} = x+2$

Therefore $\lim_{x \to 2} \frac{x^2-4}{x-2} = \lim_{x \to 2} (x+2) = 4$

So,
Function may be undefined at $x = a$, but its limit

⇒ LHL (Left hand Limit) ⇒ x approaches a from values less than a
⇒ RHL (Right hand Limit) ⇒ x " " " " more " a

- Condition for existence of limit

only if LHL = RHL ; $\lim_{x \to a} f(x)$ exists $\Leftrightarrow \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

- **Direct Substitution Method**

ex: $\lim_{x \to 3} (x^2+2x+1)$ = Put $x = 3$ $\boxed{= 16}$

- Imp. Standard Limits

$\lim_{x \to 0}$ | $\frac{\sin(x)}{x} = 1$ | $\frac{\tan x}{x} = 1$ | $\frac{x}{\sin x} = 1$ | $\frac{x}{\tan x} = 1$ | $\frac{1-\cos x}{x} = 0$ | $\frac{1-\cos x}{x^2} = \frac{1}{2}$

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$a>0, a\neq 1$

$\lim_{x\to 0}$ | $\frac{e^x-1}{x}=1$ | $\frac{\ln(1+x)}{x}=1$ | $\frac{a^x-1}{x}=\ln a$ | $\frac{(1+x)^n-1}{x}=n$ |

- Trignometric Standard Limits

$\lim_{x\to 0}$ | $\frac{\sin(ax)}{x}=a$ | $\frac{\tan ax}{x}=a$ | $\frac{1-\cos ax}{x^2}=\frac{a^2}{2}$ |

- Algebric Method (Factorization)
- Rationalization Method

} Denominator me 0 ki form nhi banani cahiye

- Limits involving infinity $\lim_{x\to\infty}\frac{1}{x^n}=0$ = { because x becomes very large $1/x$ become very small.
- L'Hospital Rule $\frac{0}{0}, \frac{\infty}{\infty}$

CONTINUITY

- A function $f(x)$ is continuous at $x=a$ if:

$$f(a)=\lim_{x\to a^-}f(x)=\lim_{x\to a^+}f(x)$$

$\boxed{1-\cos 2\theta = 2\sin^2\theta}$

$$f(a)=\lim_{h\to 0^-}(a-h)=\lim_{h\to 0^+}f(a+h)$$

ex: $f(x)=\begin{cases}2x+3, & \text{if } x\leq 2 \quad \text{LHL}\\ 2x-3, & \text{if } x>2 \quad \text{RHL}\end{cases}$

$f(2)$ ⇒ where = is there. Take opp. of =

$f(2)=2(2)+3$
$=7$

$\lim_{x\to 0^+}(2x-3)$.

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⇒ Polynomial, Log, Modulus, Trigometric ($\sin x, \cos x$), exponential } All continuous

DIFFERENTIABILITY

A function $f(x)$ is differentiable at $x=a$ if the following limit exists and is finite.

$$f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$ [called the derivative of $f(x)$ at $x=a$.]

$$f'(a)=\lim_{x\to a}\frac{f(x)-f(a)}{x-a}$$ represents instantaneous rate of change of f(x)

⇒ If a $f(x)$ is differentiable at $x=a$, then it must be continuous at $x=a$. But the reverse is not necessarily true.

- Differentiability condition

$$LHD=RHD \Rightarrow \lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h}=\lim_{h\to 0^+}\frac{f(a+h)-f(a)}{h}$$

DIFFERENTIATION

slope of a tangent

$\frac{d}{dx}$	$c=0$	$x^n=nx^{n-1}$	$ax^n=anx^{n-1}$	$e^x=e^x$	$a^x=a^x\ln a$	$\ln($
$\frac{d}{dx}$	$\sin x=\cos x$	$\cos x=-\sin x$	$\tan x=\sec^2 x$	$\cot x=-\csc^2 x$	$\sec x=\sec x\tan x$ $\csc x=-\csc x\cot$	

$\frac{d}{dx}$ | $\sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$ | $\cos^{-1}x = \frac{-1}{\sqrt{1-x^2}}$ | $\tan^{-1}x = \frac{1}{x^2+1}$ | $\cot^{-1}x = \frac{-1}{x^2+1}$

$\frac{d}{dx}$ | $\sec^{-1}x = \frac{1}{|x|\sqrt{x^2-1}}$ | $\csc^{-1}x = \frac{-1}{|x|\sqrt{x^2-1}}$

⇒ Sum Rule $y = u + v$

⇒ Product Rule : $y = uv \Rightarrow u\frac{dv}{dx} + v\frac{du}{dx}$

→ Quotient Rule: $y = \frac{u}{v} \Rightarrow \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

→ Chain Rule

• Types
- → General
- ⇒ Implicit (do both side then $\frac{dy}{dx}$ one side)
- ⇒ Parametric ($\frac{du}{dx}, \frac{dy}{dx}$ $\frac{dv}{du}$ =)
- take log both side ⇒ Logarithm (variable ki power variable) (multiple chize divide or multiply)
- → Higher - order

⇒ Rolle's Theorem : If $f(x)$ satisfies
1. $f(x)$ is conti on $[a, b]$
2. " " diffe " (a, b)
3. $f(a) = f(b)$

then there exists at least one $c \in [a, b)$ such that $f'(c) = 0$

⇒ Lagrange's Mean value Theorem: If $f(x)$ is:
1. conti on $[a, b]$
2. diffe " (a, b)

then there exists at least one $c \in (a, b)$ such that :

$f'(c) = \frac{f(b) - f(a)}{b-a}$

Diff ⇒ Conti ⇒ Limit exists
most imp example = $f(x) = |x|$

$\cos(\bar{a}b) = \cos(a+b)$

3+5 1

Q. $y = \sin(ax+b)$

$y_1 = a\cos(ax+b) = a\sin(\pi/2 + ax+b)$ $\frac{2\pi}{2}$

$y_2 = a^2\cos(\pi/2 + ax+b) = a^2\sin(\pi/2 + \pi/2 + ax+b)$

$y_3 = a^3\cos(2\pi/2 + ax+b) = a^3\sin(3\pi/2 + ax+b)$

$y_n = a^n\sin(n\pi/2 + ax+b)$

Q. $y = \cos(ax+b)$

$y_n = -a^n\cos(n\pi/2 + ax+b)$

$y_1 = -a\sin(ax+b) \Rightarrow a - \cos(\pi/2 + ax+b)$

$y_2 = a^2\cos(2\pi/2 + ax+b) \Rightarrow a^2 \sin($

Q. $y = \sin(2x)\sin 3x$
$y_n = ?$

⇒ $\sin 2x \sin 3x$ Multiply & Divide by 2

⇒ $\frac{2\sin 2x \sin 3x}{2}$ ⇒ $\sin 5x + \sin x$

⇒ $5^n\sin(\frac{n\pi}{2}) + \sin(\frac{n\pi}{2})$

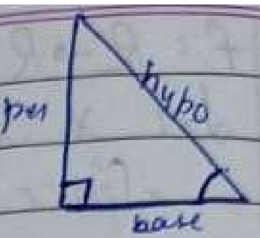
$y \Rightarrow \frac{1}{2}[\cos x - \cos 5x]$

Differentiating both side wrt 'x'

$\frac{d^n(y)}{dx^n} = \frac{1}{2}\frac{d^n}{dx^n}(\cos x - \cos 5x)$

DJF

Trignometry Class - 10



- The relationship b/w sides and Angles of a △
- It is always done in Right angle △

$H^2 = P^2 + B^2$ (PGT)

$\cot^2\theta = (\cot\theta)^2$

In this there are six Trignometric ratios

① $\sin\theta$	= P/H	0	1/2	1/√2	√3/2	1
② $\cos\theta$	= B/H	1	√3/2	1/√2	1/2	0
③ $\tan\theta$	= P/B	0	1/√3	1	√3	NOT
④ $\operatorname{cosec}\theta$	= H/P	ND	2	√2	2/√3	0
⑤ $\sec\theta$	= H/B	0	2/√3	√2	2	ND
⑥ $\cot\theta$	= B/P	ND	√3	1	1/√3	0

$\sin\theta = \frac{1}{\operatorname{cosec}\theta}$

$\tan\theta = \frac{1}{\cot\theta} = \frac{\sin\theta}{\cos\theta}$

$\cos\theta = \frac{1}{\sec\theta} =$

Same angle

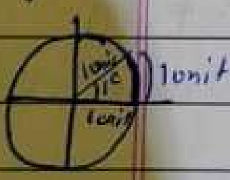
① $\sin^2\theta + \cos^2\theta = 1$ ② $\sec^2\theta - \tan^2\theta = 1$ ③ $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

Trigonometry Class - 11

(Anti-clockwise) Positive — Negative (Clockwise)

ANGLE MEASURE

Degree measure ② Radian measure (SI unit):



$\theta = \frac{l}{r}$ rad

0°	0 radian
30°	π/6 radian
45°	π/4 "
60°	π/3 "
90°	π/2 "
180°	π "

Degree to Radian = $\frac{\pi}{180}$ = 1 rad × Degree

Radian to Degree = $\frac{180}{\pi}$ × Radian

Length of Arc $l = \frac{\theta}{360} \times 2\pi r \Rightarrow l = \theta r$ (θ in degree; θ in rad)

$\sin\theta, \operatorname{cosec}\theta$	All Positive
$90°-180°$	$0 \to 90°$
$180°-270°$	$270° \to 360°$
$\tan\theta, \cot\theta$	$\cos\theta, \operatorname{cosec}\theta$

$\cos(-x) = \cos x$

$\sec(-x) = \sec x$

For this angle should be positive & in degrees

$\sin 765°$ $\quad 90\overline{)765}(8$ $\quad -72$ $\quad 45°$

$\sin(90 \times 8 + 45°)$

$\sin 45°$

$= 1/\sqrt{2}$

- first quad — +

If even no change

If odd change $\sin \leftrightarrows \cos$, $\tan \leftrightarrows \cot$, $\sec \leftrightarrows \operatorname{cosec}$

ALLIED ANGLES $\quad \sin(90-\theta) = \cos\theta$; $\cos(90-\theta) = \sin\theta$

$\tan(90-\theta) = \cot\theta$; $\cot(90-\theta) = \tan\theta$

$\sec(90-\theta) = \operatorname{cosec}\theta$; $\operatorname{cosec}(90-\theta) = \sec\theta$

42 more

QUADRANT ANGLES

IDENTITIES

$\cos(x+y) = \cos x\cos y - \sin x\sin y$; $\cos(x-y) = \cos x\cos y + \sin x\sin y$

$\sin(x+y) = \sin x\cos y + \cos x\sin y$; $\sin(x-y) = \sin x\cos y - \cos x\sin y$

$\tan(x+y) = \dfrac{\tan x+\tan y}{1-\tan x\tan y}$; $\cot(x+y) = \dfrac{\cot x\cot y-1}{\cot y+\cot x}$

$\tan(x-y) = \dfrac{\tan x-\tan y}{1+\tan x\tan y}$; $\cot(x-y) = \dfrac{\cot x\cot y+1}{\cot y-\cot x}$

$\cos 2x = \cos^2 x-\sin^2 x = 2\cos^2 x-1 = 1-2\sin^2 x = \dfrac{1-\tan^2 x}{1+\tan^2 x}$

$\sin 2x = 2\sin x\cos x = \dfrac{2\tan x}{1+\tan^2 x}$

$\tan 2x = \dfrac{2\tan x}{1-\tan^2 x}$

~~Double~~ angle Half angle

$\sin 3x = 3\sin x - 4\sin^3 x$; $\cos 3x = 4\cos^3 x - 3\cos x$; $\tan 3x = \dfrac{3\tan x-\tan^3 x}{1-3\tan^2 x}$

$\cos x+\cos y = 2\cos\dfrac{x+y}{2}\cos\dfrac{x-y}{2}$; $\cos x-\cos y = -2\sin\dfrac{x+y}{2}\sin\dfrac{x-y}{2}$

$\sin x+\sin y = 2\sin\dfrac{x+y}{2}\cos\dfrac{x-y}{2}$; $\sin x-\sin y = 2\cos\dfrac{x+y}{2}\sin\dfrac{x-y}{2}$

(i) $2\cos x\cos y = \cos(x+y) + \cos(x-y)$
(ii) $-2\sin x\sin y = \cos(x+y) - \cos(x-y)$
(iii) $2\sin x\cos y = \sin(x+y) + \sin(x-y)$
(iv) $2\cos x\sin y = \sin(x+y) - \sin(x-y)$

$\sin\frac{x}{2} = \sqrt{\frac{1-\cos x}{2}}$; $\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$; $\tan\frac{x}{2} = \sqrt{\frac{1-\cos x}{1+\cos x}}$

If x lies	then $\frac{x}{2}$ lies
I	I
II	I
III	II
IV	II

- $\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$
- $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$
- $2\tan^{-1}x = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$
  $= \tan^{-1}\left(\frac{2x}{1-x^2}\right)$
- $\sin^{-1}\theta + \cos^{-1}\theta = \frac{\pi}{2}$

ITF (Class 12)

A function is said to be invertible if it is bijective (one-one & onto)
$f(x) = \sin x \rightarrow f(0) = \sin 0 = 0$; $f(\pi) = \sin\pi = 0$ Many function - Hence not bijective.
So we restrict the domain (x) — then range becomes $[f(-x), f(x)]$

Invertible means range becomes domain and domain becomes range.

	Domain	Range		Domain	Range
$\sin x$			$\sin^{-1}x$	$[-1, 1]$	$[-\pi/2, \pi/2]$
$\cos x$			$\cos^{-1}x$	$[-1, 1]$	$[0, \pi]$
$\tan x$			$\tan^{-1}x$	R	$(-\pi/2, \pi/2)$
$\cot x$			$\cot^{-1}x$	R	$(0, \pi)$
$\sec x$			$\sec^{-1}x$	$R-(-1,1)$	$[0,\pi]-(\frac{\pi}{2})$
$\operatorname{cosec} x$			$\operatorname{cosec}^{-1}x$	$R-(-1,1)$	$[0, \pi]$

Graphs

Some important formulas

- $\sin^{-1}(\sin x)$ or $\sin(\sin^{-1}x) \rightarrow x \in [-\pi/2, \pi/2]$
- $\sin^{-1}(-x) = -\sin^{-1}x$; $\tan^{-1}(-x) = -\tan^{-1}x$; $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$
- $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$; $\cot^{-1}(-x) = \pi - \cot^{-1}(x)$; $\sec^{-1}(-x) = \pi - \sec^{-1}(x)$